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Multi-UAV path planning utilizing the PGA algorithm for terrestrial IoT sensor network under ISAC framework

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Abstract

The integration of internet of things (IoT) technologies with unmanned aerial vehicles (UAVs) has initiated a groundbreaking revolution in data acquisition and communication systems across diverse domains. This document introduces an innovative endeavor, the integration of multi-UAV path planning for integrated sensing and communication (ISAC) within ground-based CAT-M1 IoT sensor networks, accomplished through the application of the Pareto-based genetic ant colony optimization (PGA) algorithm. The PGA algorithm is highly capable of UAV path planning due to its efficiency, adaptability, and seamless integration of domain expertise. By employing the PGA algorithm, we simultaneously minimize UAV travel distance while optimizing energy consumption, resulting in multi-objective optimization. The synergy of ground-based IoT sensors and seamless UAV communication, coupled with a convex optimization resource allocation algorithm, empowers real-time data acquisition and heightens situational awareness. Our proposed UAV path planning PGA algorithm with resource allocation is crafted to maximize the efficiency of ground-based sensor data acquisition. We stand at the forefront of advancing multi-UAV data collection systems with this pioneering approach, promoting increased efficiency, robustness, and transformative solutions across diverse domains. The proposed system for ISAC achieves an impressive throughput of up to 95% of the system's capacity.

1 | INTRODUCTION

UAVs and IoT technologies have transformed data acquisition and communication systems in a variety of domains.¹ The landscape of data acquisition and communication has undergone a profound shift due to the alliance of UAVs and IoT technologies.² The integration of UAVs into data sensing and communication, forming the ISAC framework, emerges as a transformative solution for swift and effective information gathering within ground-based IoT sensor networks.³ This paper introduces the PGA algorithm to tackle the intricate challenges of optimizing UAV trajectories for data collection from distributed sensors within the ISAC framework. The algorithm not only ensures seamless communication and dependable data transmission but also introduces the notion of resource allocation, a pivotal aspect in achieving efficiency and maximizing the data collection system performance. By synergizing the strengths of pareto-based optimization,⁴ genetic algorithms (GA)⁵ and ant colony optimization (ACO),⁶ the PGA Algorithm formulates path planning as a multi-objective optimization task, embracing ISAC's resource allocation⁷ in its core. The algorithm's objectives encompass

reducing overall UAV travel distance, maximizing sensor coverage, optimizing energy consumption, and adhering to ISAC's communication range and connectivity constraints. This approach yields a diverse set of pareto-optimal paths, harnessing Pareto-based optimization, genetic evolution, and ant colony behavior, enabling intelligent decisions tailored to ISAC's mission objectives.

The SVM-ECN technique for classifying nodes in IoT networks is introduced in efficient selfish node detection using SVM in IoT-MANET Environment.⁸ This method improves network efficiency by separating possible attackers and evaluating historical behavior for trustworthiness. It is suggested to use this technique to protect the network from malevolent nodes. The ETTS algorithm is proposed for UAV task scheduling, with an emphasis on efficient energy utilization and reduced latency in IoT systems via clustering, ARMA model predictions, and multi-objective optimization.⁹ This technique is proposed for optimizing job scheduling for UAVs in IoT systems, with a focus on energy efficiency and minimizing delays. Investigates the revolutionary influence of UAVs and the IoT, offering green UAV-based fog computing for energy-efficient data computation and communication in crucial applications.¹⁰ This novel technology is proposed to revolutionize data processing and communication in essential applications, with an emphasis on the environmentally benign elements of UAV-based fog computing. Using hexagonal cell clustering and the ARMA model for trajectory planning, energy and delay optimized trajectory planning for UAV-IoT environments tackles issues in multi-UAV multi-IoT networks, resulting in significant time and energy savings.¹¹ This method is suggested to address difficulties in multi-UAV and multi-Internet of Things scenarios. It emphasizes the use of ARMA model and hexagonal cell clustering for effective trajectory planning, which eventually results in significant energy savings and shorter processing times.

The importance of UAVs in disaster management, proposing a layered architecture considering 3D parameters for improved capacity, coverage, and network performance in urban disasters.¹² This emphasis on a layered architecture is suggested to enhance the overall capabilities, coverage, and network performance of UAVs in disaster management, particularly in urban settings. Examines UAV application security applications, emphasizing the use of machine learning, blockchain, and watermarking to safeguard data from hackers and unauthorized actions.¹³ This analysis emphasizes the importance of using modern security techniques like machine learning, blockchain, and watermarking to protect UAV data from potential threats and unwanted actions. Utilizing unmanned aerial vehicles (UAVs), a HetNet wireless network was implemented to address urban data traffic issues. The results showed increases in spectral efficiency, transmission range, and overall network performance in highly populated locations.¹⁴ This strategy is put forth to address issues with urban data traffic and highlights the benefits of a HetNet wireless network enabled by UAVs, which lead to increased spectral efficiency, increased transmission range, and improved network performance in densely populated metropolitan regions.

This paper distinguishes itself by offering an integrated solution for multi-UAV path planning in ISAC, addressing both sensing and communication challenges simultaneously. The multi-objective optimization using the PGA algorithm sets apart in terms of efficiency and adaptability across various dynamic scenarios.

Numerous artificial intelligence and machine learning algorithms are at your disposal for diverse tasks involving UAVs. Nevertheless, opting for the PGA algorithm seems warranted given the problem's inherent qualities. This choice underscores the significance of interpretability, computational efficiency, problem-specific adaptability, robustness in simulation, and the seamless integration of domain expertise. Despite the merits of AI and machine learning, the distinctive features of the optimization problem render traditional methods a fitting and efficient option. Main contribution of this paper is:

1. We elaborate the detail formulation and execution of the PGA algorithm, extending its scope to encompass multi-UAV path planning within the ISAC framework, infused with resource allocation capabilities.
2. We evaluate the system's performance through the simulations, showcasing the efficiency, robustness, and adaptability of the proposed ISAC-based approach in diverse scenarios.
3. We highlight ISAC's significant contributions to multi-UAV path planning, highlighting transformative applications.
4. We integrate UAV-based data collection systems by presenting the novel ISAC driven approach, which improves the data acquisition and communication efficiency of IoT and UAV technologies.

The remaining portions of the document are organized as follows: Section 2 provides an overview of the related works. Section 3 presents an overview of ISAC and the proposed PGA algorithm with system model, problem formulation, and proposed PGA algorithm for ISAC explained in Section 4. The result discussion, conclusions and future work are covered in Section 5, 6, and 7 respectively.

2 | RELATED WORKS

A multitude of algorithms and strategies have undergone development. The Dijkstra algorithm⁸ specializes in determining the shortest path through graph nodes. As one of the most widely adopted search algorithms for route planning, the search algorithm⁹ bolsters performance by integrating heuristics to navigate the search process, particularly in two-dimensional scenarios. Recent progress has expanded the utility of these techniques to encompass three-dimensional (3D) geographical environments.

Kanchan Vally et al.,¹⁰ along with fellow researchers,¹¹ confronted the challenge of 3D path planning for UAV navigation within environments replete with stationary obstacles. Nikolos and Valavanis¹² harnessed B-spline curves to model an aircraft's 3D flight path, optimizing the control points using an evolutionary algorithm. However, obstacle avoidance remains a formidable hurdle in this context. The development of intelligent optimization algorithms plays a pivotal role in addressing the intricate quandaries of route determination, rooted in diverse objective functions. For instance, within underwater domains, evolutionary algorithms employing binary coding have been employed to tackle 3D path-finding challenges for underwater vehicles, where paths are delineated as a sequence of cells within a 3D grid. Doherty et al.¹³ innovated a technique to maneuver a quad-rotor vehicle into a stable hover position.

UAVs possess the capacity to ascend vertically and sustain stable hover positions, rendering them well-suited for scanning missions. The fuzzy particle swarm optimization (PSO) method has been instrumental in maintaining a fixed altitude for precise scanning. Tulum et al.¹⁴ introduced an agent-based solution for UAV mission route planning, factoring in considerations such as targets, threats, terrain, and airspace restrictions when ascertaining the optimal array of waypoints for UAV missions.

An additional study¹⁵ explores regions rich in obstacles, offering an overview image and scrutinizing deterministic and probabilistic path planning strategies for self-reliant UAV networks. The study presents algorithms amenable to both online and offline implementation, underscoring the approach's efficiency with a reduced number of images. Nevertheless, this approach, unlike probabilistic methodologies that offer greater adaptability and flexibility, necessitates heightened knowledge and time for generation. A comprehensive evaluation of UAV-based value-added IoT services is featured in Reference 16, centering on UAV collision avoidance strategies, challenges in wireless communication within flying ad-hoc networks (FANET), the utilization of on-board UAV sensors, and the processing of pertinent data amassed by UAVs.

The sensor network with mobile agents (SENMA) design,¹⁷ championed by L. Tong et al., introduces extensive, low-power sensor networks employing mobile agents. These mobile agents serve as the receiving terminals in SENMA, conferring substantial energy efficiency advantages, particularly in flat ad hoc network settings. Ho et al.¹⁸ delve into the exploitation of UAVs for data collection and the selection of sensor network communication topologies. Their principal aims encompass the minimization of travel time, the maximization of communication quality, and the conservation of energy. A groundbreaking bio-inspired multi-objective optimization algorithm, denominated MOCAS,¹⁹ is put forward for wireless sensor networks, incorporating metrics such as generational distance, error ratio, spacing, and more. Challenges persist in configuring algorithm parameters and managing constrained problems within this domain. To guarantee seamless connectivity between individuals and objects, along with the enhancement of particle multi-swarm optimization performance, a SWARM-based data delivery algorithm²⁰ is proffered.

In Reference 15 STSQL introduces a method to optimize the paths of UAVs in unfamiliar terrains. Utilizing a spatial and temporal substate-based Q-learning algorithm. In Reference 16, the authors suggested UAV path planning through a combination of A* and Q-learning techniques, considering the safety and security constraints of the UAV. Another study,¹⁷ delved into planning for multiple UAVs in flying ad-hoc networks (FANET) by employing data routing. The authors enhanced the UAV network link quality by incorporating a Q-learning-based fuzzy logic algorithm, considering the mobility of the UAVs. Additionally, Chen et al.¹⁸ introduced a cluster-based coverage path planning approach for diverse UAVs with varying flying speeds and energy supplies.

The deep reinforcement learning (DRL) technique has recently become a go-to for UAV trajectory planning in data collection scenarios. In Reference 19, an approach using DRL was introduced for CH (Cluster Head) selection in dynamic networks. Moreover Reference 20, explored an energy-minimizing and clustering approach for sequential trajectory planning, showcasing the effectiveness of DRL in single UAV operations. The authors of²¹ put forth a DRL-based platform for data gathering with multiple UAVs. In Reference 22, energy limitations and sensing information update scenarios for data gathering with multiple UAVs were considered, employing a deep Q-network and showcasing promising performance. Li et al.²³ suggested a DRL algorithm for charging ground devices and data collection, aiming to minimize overall data packet loss. The algorithm factored in battery levels and the data queue length of sensors during UAV visits to predefined

waypoints. It's important to note that, in general, deep learning algorithms exhibit higher computational complexity than the Q-learning technique, with a relatively lower convergence speed compared to tabular-based Q-learning.

Efficient resource allocation in cooperative cognitive radio networks is addressed by Muhammad Naeem,⁷ discussing the taxonomy of objectives and protocols used in the literature for this purpose. S. Giannecchini proposes a fast online resource allocation algorithm (CoRAL) for dynamically reconfiguring sensor networks in response to new hot spots.²⁴

3 | OVERVIEW OF ISAC

With the development of mobile intelligent applications, for example, auto driving, indoor and outdoor location, sensing has played a more important role than ever before, which motivates the joint design of communication and sensing.²⁵ ISAC is an emerging technique that utilizes the same spectrum resource to realize both communication and radar sensing functions.²⁶ ISAC merges sensing and communication functions in a single system, enhancing efficiency and reducing interference. The ISAC system manages interference and resource allocation. They use advanced communication protocols and frequencies, like terahertz and 5G, to facilitate tasks like vehicle-to-vehicle communication and localization. The 3GPP focuses on ISAC standards for future releases. In practical applications, ISAC connects ground based IoT sensors with UAVs through a robust infrastructure. This integration maximizes the effectiveness of IoT-enabled data collection and transmission.

3.1 | Terrestrial IoT for ISAC

ISAC systems operated separate sensing and communications (S&C) components, utilizing distinct hardware within the same frequency bands, and employing techniques to prevent mutual interference. However, as S&C operations increasingly merge, the need for integrated hardware arises. ISAC systems are now broadly categorized into two types: radar-communications coexistence (RCC) and dual-functional radar-communications (DFRC). RCC aims to manage interference and resource allocation, enabling both systems to function without disrupting each other. On the other hand, DFRC focuses on executing S&C tasks on a shared infrastructure.²⁷

While established millimeter-wave (mm-Wave) communication protocols and waveforms, such as the IEEE 802.11ad Wi-Fi standard, have been suggested for aiding vehicular sensing through communications, recent research has applied similar signaling methods to facilitate vehicle-to-vehicle (V2V) ISAC in the lower terahertz (THz) range (.06–4 THz). The third-generation partnership project (3GPP) Release-16 has outlined specifications for 5G localization and sensing in monostatic mode using time difference-of-arrival (TDoA) techniques.²⁸ Notably, a current work item (identified by reference number S1-220144) within 3GPP is dedicated to ISAC.

The ground based IoT sensor network comprises numerous sensors distributed across three distinct sensing zones. Within each zone, ten IoT sensors are positioned at random to monitor and gather data from the surrounding environment. These sensors capture data and then transmit it to the UAVs for further processing and communication. To ensure smooth communication between the UAVs and the IoT sensors, a robust communication infrastructure has been established. This infrastructure enables dependable and efficient data transmission through IoT technology, which offers low power, wide-area connectivity for IoT devices. The communication infrastructure facilitates data exchange between the UAVs and the ground-based sensors. The proposed system architecture utilizes the PGA algorithm to optimize the UAVs' paths, ensuring efficient coverage and reduced energy consumption during data collection. Furthermore, the integration of IoT technology guarantees reliable and low-latency communication between the UAVs and the ground-based sensors, enabling timely data transmission.

In the subsequent sections of this paper, we will delve into further detail regarding the implementation of the PGA algorithm, the communication system design, and the performance evaluation metrics, showcasing the effectiveness and benefits of our proposed system architecture. In this scenario, the resource allocation and management are addressed by using a Pareto-based genetic algorithm with an ant colony optimization algorithm to plan the paths for the three UAVs. The objective is to optimize the paths such that data can be efficiently collected from the randomly placed IoT sensors in the three sensing zones.³

Resource allocation refers to the efficient distribution and utilization of resources like energy, bandwidth, time slots, and processing power among the various tasks in the system. In the scenario, the resources of the UAVs (e.g.,

battery power, communication bandwidth) need to be allocated to both by sensing the sensor and communication (data transmission) functions.²⁹

By optimizing the paths of the UAVs using the mentioned algorithms, aiming to maximize the data collection efficiency while minimizing the communication overhead and latency. The chosen paths will determine how efficiently the UAVs can communicate with the IoT ground sensors and transmit the collected data. The optimized paths can help avoid collisions, interference, and unnecessary resource consumption during communication.

3.2 | Components and network architecture for ISAC

Resource allocation refers to the efficient distribution and utilization of resources like energy, bandwidth, time slots, and processing power among the various tasks in the system. In the scenario, the resources of the UAVs (e.g., battery power, communication bandwidth) need to be allocated to both by sensing the sensor and communication (data transmission) functions.²⁹

The data transmission and communication system design play a vital role in facilitating seamless and reliable communication between the UAVs and the ground based IoT sensors in the context of the multi-UAV data collection system. The design encompasses the integration of sensing and communication materials to ensure efficient data transmission, low latency, and robust connectivity for data exchange. Sensors transform physical information into digital signals via analog-to-digital converters (ADCs) and receive signal conditioning, overseen by microcontrollers. The setup incorporates memory, power management, interfaces, and protective housing for dependable data acquisition. The integrated communication system encompasses IoT components such as modems, antennas, and SIM cards, facilitating secure wireless data transmission with power efficiency. This architecture enables effortless integration and real-time insights within IoT ecosystems.³⁰

3.2.1 | Components for UAV-aided ISAC

The ground based IoT sensor network consists of sensors distributed across three sensing zones, measuring 30×30 . These sensors are equipped with various sensing materials, including environmental sensors, proximity sensors, or specialized sensors tailored to the specific application requirements. These sensing elements enable the sensors to capture and measure relevant environmental data such as temperature, humidity, air quality, or domain-specific parameters. Sensors detect physical phenomena, converting them into electrical signals. ADCs transform analog signals into digital data.²⁵ Signal conditioning enhances accuracy, and microcontrollers manage processing and control. Memory stores data and firmware, while power management ensures efficient energy use. Interfaces enable communication with other components, and calibration circuitry maintains accuracy. Housing safeguards components from environmental factors. This integrated setup forms the basis for collecting reliable data for transmission and analysis within the system.

The communication part of an integrated system with an IoT sensor involves specialized components like the IoT module, tailored antenna, SIM card, and transceiver. These elements enable wireless data transmission over the IoT network. Protocol stacks manage communication layers, while security features ensure data privacy. Power management optimizes energy use, and network management interfaces with cellular infrastructure. Cloud connectivity facilitates data exchange, and firmware updates can be managed remotely. Diagnostic tools monitor communication health. Together, these components enable efficient, secure, and reliable data transmission from the sensor to remote platforms, enriching IoT ecosystems with seamless integration and real-time insights. A robust framework architecture of ISAC is shown in Figure 1. The ISAC architecture operates across three distinct levels: terminal, edge, and cloud. At the terminal level, various scenarios, including intelligent vehicle networks, intelligent manufacturing, airborne networks, and seaborne networks, seamlessly join sensing communication and computation.³¹ These scenarios enable data communication and transmission. The data is collected at this level and then transmitted to the edge level, where a base station employs beamforming (BF) to gather the data. The edge level acts as a bridge, forwarding the collected information to the cloud level as ISAC data. In the cloud, intelligent computing processes the data and sends the results back to the previous layers, completing the ISAC architecture's data flow cycle.

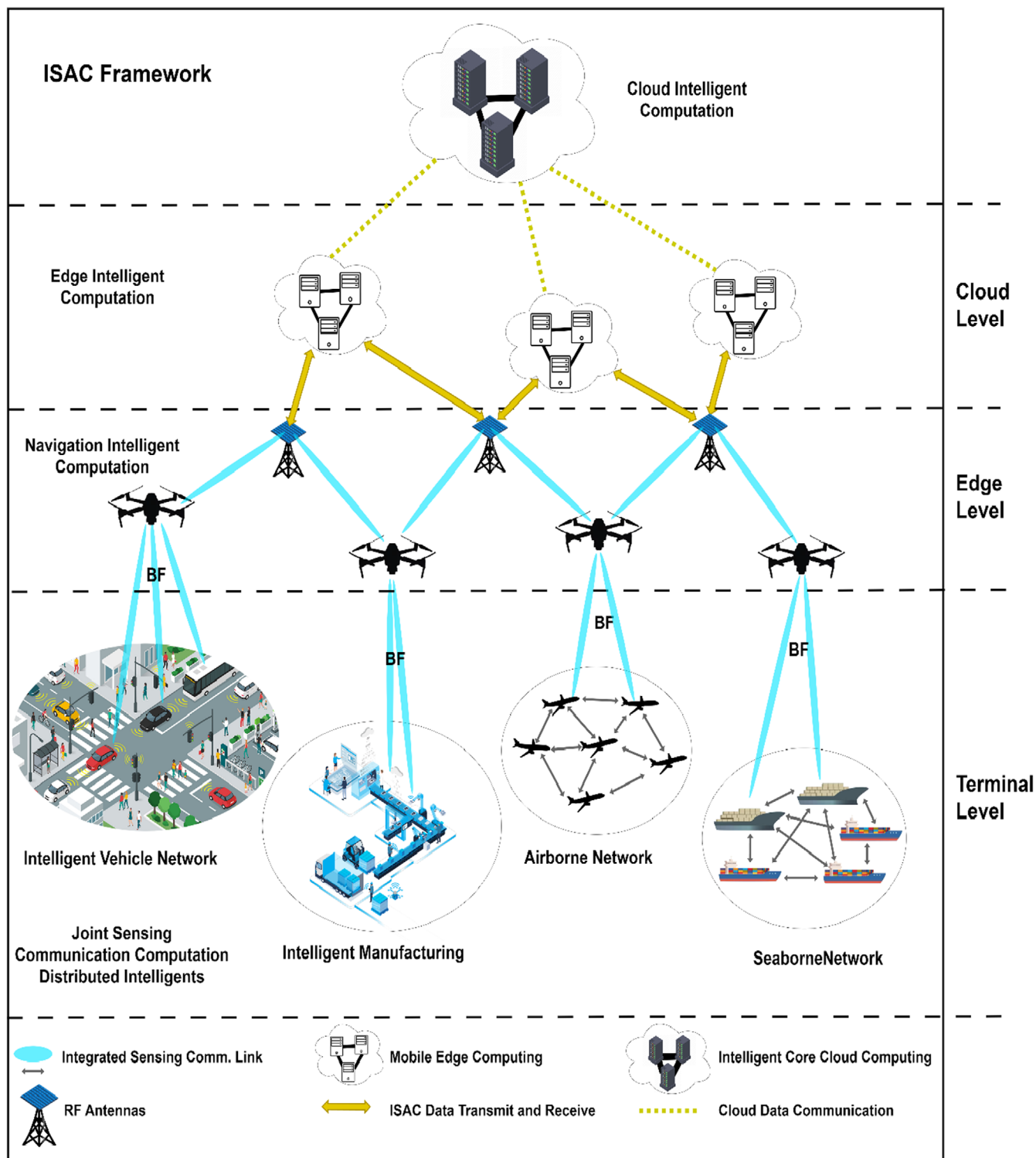


FIGURE 1 Illustration of UAV-aided ISAC.

3.2.2 | Network architecture for ISAC

The ISAC network architecture comprises sensors for data collection, data processing and conversion components, micro-controllers, memory, power management, interfaces, protective housing, IoT communication elements, and associated protocol stacks, security features, power optimization, and network management. This architecture enables efficient, secure, and reliable data collection, transmission, and integration within IoT ecosystems. The data transmission process begins with the UAVs collecting data from the ground-based sensors. The sensors, equipped with sensing materials, capture and process environmental data, converting it into digital information for transmission. Using the optimized paths, the UAVs efficiently navigate the sensing zones, minimizing the distance traveled for data collection.² The UAVs establish direct communication links with nearby sensors through IoT communication modules. To enhance system performance, interference coordination techniques are implemented. These techniques guarantee that concurrent transmissions from multiple UAVs and sensors do not disrupt one another, thereby minimizing the adverse effects on signal quality and the overall system's performance. The interference coordination mechanism oversees various aspects such as frequency channels, transmission power levels, or time slots, optimizing the allocation of resources and effectively mitigating interference effects.

4 | PROPOSED PGA ALGORITHM

4.1 | System model

We built an environment where sensors are distributed in the sensing area in a random manner. The collected data is transmitted from the sensors to the UAVs in real time or periodic intervals, depending on the application requirements. The UAVs receive the data, perform preprocessing or compression as necessary, and transmit it to the designated data storage or processing infrastructure. Figure 1 shows the sensing and communication between the UAV and sensors with multiple scenarios namely vehicle networks, manufacturing, airborne networks, and seaborne networks. The UAV moves towards a sensor and sends wakeup signal. After receiving the wakeup signal sensor prepares for the transmission and sends a signal to the UAV. Communication developed between UAV and sensor. When communication is developed, the sensor starts transmitting the sensing data to UAV.

The work leverages the PGA algorithm to optimize the paths of the UAVs, ensuring efficient coverage and reduced energy consumption during data collection and transmit data by uplink communication. This is achieved through the path planning using pareto based GA with ACO as illustrated in Figure 2. This model incorporates all the sensors of a sensing area and serializes all the sensors using GA. We assume by finding the serial of sensor give us the opportunity to assume the primary path plan to cover the full sensing zone to fly through the sensing area in a cost effective and collision freeway. After getting the best serial we find the best trajectory between sensors in the efficient way. ACO helps us to find the best trajectory between sensors. With the multiple iteration we made multiple versions of the path and made a comparison between them and chose the best one by considering fitness function, objective, energy consumption, flight risk type multiple constraint.

Resource allocation algorithm for ISAC using convex optimization revolves around optimizing convex objective functions under the presence of linear or convex constraints. The procedure encompasses formulating the optimization challenge alongside its constraints, opting for suitable convex functions, and deciding on optimization variables. Convex optimization resource allocation algorithm³² find utility in various domains for the effective distribution of resources for sensing and communication. They ensure optimal outcomes and come with assured convergence characteristics.

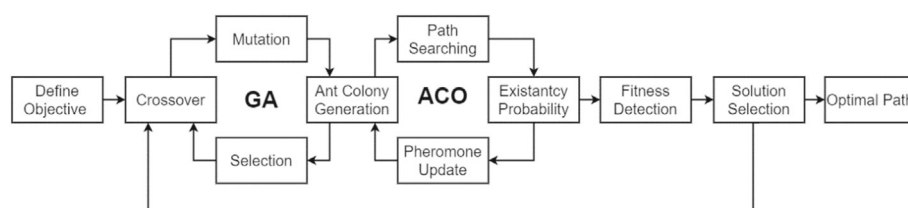


FIGURE 2 Block diagram of the PGA algorithm.

4.2 | Problem formulation

Path planning in the context of the multi-UAV data collection and uplink data transmission system can be formulated as an optimization problem with specific objectives and constraints.³³ The goal is to find the optimal or near-optimal paths for each UAV to traverse the sensing zones and collect data from the IoT sensors while considering various factors that affect the system's performance. The objective is to minimize the total distance traveled by the UAVs while maximizing sensor coverage and optimizing energy consumption.³⁴ By solving the path planning problem using the optimization algorithm, the proposed approach aims to generate a set of pareto-optimal paths for the UAVs, offering various options for the system designer to choose from based on their priorities and preferences. The final selected paths will be used to guide the UAVs in collecting data from the IoT sensors efficiently and effectively, while considering system constraints and ensuring reliable communication performance to transmit the data. Within our operational scenario, characterized by a 3D operational space, numerous sensors are dispersed randomly throughout the sensing area. These sensors are tasked with devising flight paths for UAVs possessing comparable specifications. During the path planning we impose that the UAVs avoid each other and stay in their sensing zone and find the sensors. Furthermore, the communication between UAVs is done with each other by UAV-to-sensor communication system. In our execution, we divided the entire operational space into several small unit areas referred to as cells, with certain cells designated as restricted zones. The depiction of the operational area and the trajectory planning approach we propose are elaborated upon in the subsequent sections.

The described process involves a two-step optimization approach for UAV path planning. Initially, a genetic algorithm determines the best order in which to visit sensors, aiming to minimize travel distance. Then, an ant colony optimization method refines this sequence by using pheromone trails and heuristics to guide artificial ants along potential paths. The iterative process continues until convergence or a stopping criterion is met, combining exploration and exploitation to discover optimal or near-optimal UAV paths that efficiently cover sensors. The integrated PGA algorithm considers multiple objectives, such as distance, sensor coverage, and energy consumption, using a Pareto-based approach to provide diverse and optimal solutions. This integration enhances data collection efficiency from ground based IoT sensors, reducing redundant movements, and improving overall system performance.

Using the optimization approach, we built trajectory in 3D space from a sequence of waypoints. As a result, vector form is employed for encoding feasible paths, with the vector element $P_i = (a_i, b_i, c_i)$ representing the i th waypoint,³⁵ as shown below:

$$P_r = P_1, P_2, P_3, \dots, P_{N_t} \quad (1)$$

where N_t is known as the number of trajectories in a feasible path.

In this part, multi-objective fitness function made up of the trajectories produced by the multi-UAV path planning method we use. We divided the indexes for optimization into two groups and gave each group a different priority level. This is done in order to emphasize the various weights that optimization indexes carry during the optimization process. They can be divided into two categories: (1) the constraints that UAVs must adhere to because of their physical limitations, and (2) the optimization objectives that must be maximized in order to generate an ideal trajectory. As a result, we define the fitness function as:

$$FT_{fitness} = FT_{obj} + FT_{const} \quad (2)$$

where FT_{obj} is the objective function, concentrating on maximizing values in terms of other parameters, and FT_{const} is the physical and environmental constraints, should be completed prior to trajectory planning.

The objectives of the function are delineated as an optimization criterion aimed at enhancing the quality of path planning.¹ Consequently, we define the objective function as a composite of energy consumption and flight risk estimation, and we express it in the following manner, as in (3):

$$FT_{obj} = -wt_1 FT_{CE} - wt_2 FT_{RF} \quad (3)$$

where wt_1 and wt_2 were utilized to describe the weight of the objective component and consumption of energy (FT_{CE}) and risk of flying (FT_{RF}) are defined to be in the range $[0,1]$. The creation of the ideal path with the lowest energy consumption,

lowest environmental flight risk, and highest surveillance was our main goal. As a result, we designate the relevance of the monitoring region as the positive value and the other values as the negative.³⁵

Due to the inherent physical limitations of small UAVs, notably battery power constraints, meticulous path planning is imperative. It ensures that the UAV can effectively survey its designated route while making optimal use of its limited battery capacity. Consequently, prioritizing routes with lower energy consumption is consistently favored. Assuming a constant velocity is maintained during UAV operation, we proceeded to calculate the energy consumption,³⁵ CE , as follows:

$$CE = \frac{\sum_{i=1}^{N_w-1} CE_i}{CE_{max}} \quad (4)$$

where individual consumption of energy CE_i is calculated as:

$$CE_i = PO_w \times t_{i,i+1} \quad (5)$$

$$t_{i,i+1} = \frac{dt_{i,i+1}}{v} \quad (6)$$

$$dt_{i,i+1} = \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2} \quad (7)$$

where CE_i the amount of energy used from waypoint i to waypoint $(i + 1)$. PO_w is the energy used by the UAV at velocity for a unit of time, and $t_{i,i+1}$ is the amount of time it takes the UAV to fly between waypoints i th and $(i + 1)$ th. The distance in Cartesian coordinates between the sensors i th and $(i + 1)$ th waypoint is known as $dt_{i,i+1}$. We calculate the CE_{max} after normalizing the energy consumption:

$$CE_{max} = (N_w - 1) \times PO_w \times \frac{dt_{max}}{v} \quad (8)$$

where waypoint is known as dt , maximum waypoint is generated to find the best, and the maximum dt can be calculated as:

$$dt_{max} = \sqrt{X^2 + Y^2 + Z^2} \quad (9)$$

where the 3D environment's X , Y , and Z boundaries de. Furthermore, flying at a high altitude might be very risky because the wind is greater and could increase the possibility of being accidentally destroyed fine the operation space.³⁵

There are several physical restrictions that must be followed throughout flight time since they make the UAV more susceptible to inclement weather (such as line rain, strong winds, or snow, etc.) when performing a surveillance duty. Two types of flight risks are established based on the situation:

- Environmental risk
- High altitude risk

Estimating the risk of flying can be represented RF as follows:

$$RF = \frac{\sum_{i=1}^{N_w-1} RF_i}{maxRF} \quad (10)$$

$$RF_i = \varphi_{ER} r_{i,i+1}^e + \varphi_{HAR} r_{i,i+1}^{ha} \quad (11)$$

where φ_{HAR} stands for high altitude risk and φ_{ER} for environmental risk, RF_i is the flight risk prediction can be calculate from the i th waypoint to the $(i + 1)$ th waypoint. We normalized the estimated flight risk and came up with the formula max FRE³⁵:

$$maxRF = (N_w - 1) \times [\varphi_{HAR} \times Z + \varphi_{ER} \times (2maxr^e)] \quad (12)$$

To determine whether the resulting path was feasible, a negative constraint function was created. Each restriction is set to zero when the requirements are met, and a negative penalty value P is applied otherwise. Below is a description of the constraint function. The constraint function is formulated as follows:

$$FT_{const} = TAC + OAC + CRC + CA \quad (13)$$

where consider the turning angle constraint (TAC), operational area constraint (OAC), coverage range constraint (CRC), and collision avoidance (CA) as the following constraints as FT_{const} . The UAV's flight path must offer sufficient smoothness for easy maneuverability. Consequently, the UAV's turning angle must remain below the maximum allowable turning angle. This constraint can be expressed as TAC in the following manner:

$$TAC = \sum_{i=1}^{N_w} TAC_i \quad (14)$$

where individual turning angle constraint TAC_i calculated as:

$$TAC_i = \begin{cases} P, & \text{if } \theta > \theta_{max} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where θ defines the turning angle of the UAV in 3-D directions (x_i, y_i, z_i) , and θ_{max} is the maximum tolerable turning angle. The OAC of a mission can be formulated as follows:

$$OAC = \sum_{i=1}^{N_w} OAC_i \quad (16)$$

where individual operational area constraint OAC_i calculated as:

$$OAC_i = \begin{cases} P, & \text{if waypoint in map}(x_j, y_j) \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

$$Range(x_j, y_j) = \{l_x \leq x_j \leq u_x\} \cap \{l_y \leq y_j \leq u_y\}$$

where N_w is the number of waypoints, l_x, l_y is the lower bounds of x and y , u_x, u_y is the upper bounds of x and y respectively.

For whole coverage, UAV needs to cover all the areas. If the path planning includes the coverage area, no penalty is given. The Coverage range constraint can be formulated as:

$$CRC = \sum_{i=1}^{N_w} CRC_i \quad (18)$$

where individual coverage range constraint CRC_i calculated as follows:

$$CRC_i = \begin{cases} P, & \text{if waypoint } i \text{ in Range}(x_j, y_j) \\ 0, & \text{otherwise} \end{cases} \quad (19)$$

$$range(x_j, y_j) = \{l_x \leq x_j \leq u_x\} \cap \{l_y \leq y_j \leq u_y\}$$

where N_w is the number of waypoints, l_x, l_y is the lower bounds of x and y , u_x, u_y is the upper bounds of x and y .

The meticulous design of UAV paths to eliminate collisions among them is regarded as one of the pivotal tasks in crafting viable path planning. In the case of separate trajectories, it is imperative to maintain a minimum distance between UAVs to avert collisions. Consequently, collision avoidance constraints can be delineated as:

$$CA = \sum_{i=1}^{N_w^p} \sum_{j=1}^{N_w^p} CA_i \quad (20)$$

where avoidance of collision constraint CA_i formulated as:

$$CA_i = \begin{cases} P, & \text{if } d_{ij}^{mn} < d_{min} \\ 0, & \text{otherwise} \end{cases} \quad (21)$$

where d_{min} is the minimum safe distance to avoid collision, d_{ij}^{mn} is the Cartesian distance between the i th waypoint of p th UAV trajectory and the j th waypoint of q th UAV trajectory.

4.2.1 | Genetic algorithm

There are N individuals in the population which can be regarded as a solution of a path selection scheme for the multi-UAV system. Each UAV selects randomly N paths $\{l_i^1, l_i^2, \dots, l_i^N\}$ so all the UAVs construct the population $\{ind_1, ind_2, \dots, ind_N\}$. Fitness functions in centralized GA global information to evaluate global solutions. The fitness function in distributed GA can only make use of local knowledge. The former enables UAVs to be prone to paths with greater benefits while the latter helps them to escape from local optimal solutions. The fitness of l_i^N is expressed³⁷ as follows:

$$fit_i^n = 1 - \frac{f_i(l_i^n)^{\log(t_i(a, cGA))}}{\sum_{n'=1}^N f_i(l_i^{n'})^{\log(t_i(a', cGA))}} \quad (22)$$

$$t_i(a, cGA) = t_i(a, cGA - 1) + \sum_{n'=1}^N \langle l_i^{n'} = a \rangle$$

where $f_i(l_i^n)$ represents the individual objective function return value corresponding to l_i^n ; a and a' are values; cGA is the current number of iterations of GA and an iteration represents an execution of all genetic operators; $\langle l_i^{n'} = a \rangle$ is an indicator function returning to 1 if $l_i^{n'}$ equals to a ; $t_i(a', cGA)$ is the number of times that a' is chosen by crossover point U_i until cGA , and the larger the value of $t_i(a', cGA)$ is, the more likely a' is to be local optimal. U_i evaluates its paths according to the fitness function. To select individuals for crossover, the roulette wheel selection operator is introduced to generate the crossover list $crossList_i$ composed of N indexes for U_i , and before this, a survival probability p_i^n is assigned to l_i^n ³⁴:

$$p_i^n = \frac{fit_i^n}{\sum_{n'=1}^N fit_i^{n'}} \quad (23)$$

In the crossover point U_i rearranges its values in the order of $crossList_i$ with probability p_{cross} to generate new individuals. p_{cross} is designed as follows:

$$p_{cross} = \frac{\log(mGA - cGA)}{\sqrt{mGA}} \quad (24)$$

where mGA is the maximal number of iterations of GA.

In the selection and crossover operators, values with high fitness are more likely to be preserved. They will gradually occupy the majority of the population, leading to a decline in population diversity.

$$p_{mut}^{i,n} = \frac{fit_i^n \times e^{-\frac{(cGA - \frac{1}{4}mGA)^2}{mGA}} \times (mGA - cGA)}{mGA} \quad (25)$$

To slow down the tendency, values with high fitness are made more likely to change in the mutation operation, so in Equation (25) the mutation probability p_{mut} is positively correlated with the fitness fit .

4.2.2 | Mathematical model of ACO

The ACO is a bionic algorithm that draws inspiration from how ants forage in the wild. An ant colony can always find the best route from the ant nest to a food source when it is foraging. Normally, as they travel in search of food, ants release pheromones that other ants can recognize as belonging to the same species. The remaining pheromone concentration is higher on the shorter path because, in most cases, the pheromone will evaporate over time. As a result, the ant swarm can choose the best route and tends to move in the direction of strong pheromone signals.

The first step is initializing ACO's parameters. These variables include the maximum number of iterations K , the total number of ants M , the expected heuristic factor α , the pheromone evaporation factor β , and the pheromone intensity value Q . Additionally, it is necessary to initialize the path routing memory matrix, the heuristic information matrix, and the pheromone value in the same way. One of the crucial ACO steps is learning how to select the best course of action. The roulette wheel selection is used in traditional ACO. According to this method, each candidate's likelihood of selection is inversely correlated with their fitness. In conventional ACO, Equation (26) is defining the probability selection rule for an ant moving from the i th point to the j th point for the k th ant⁶:

$$pb[i,j] = \frac{((T[C_l][i,j])^\alpha) \times (H[C_l[i,j]^\beta])}{S_{pb}} \quad (26)$$

in which summation of probability (S_{pb}), probability (pb). On the other hand, to calculate the best path between recent sensor and the next sensor need to update the pheromone matrix T :

$$\delta T[i][j] = \frac{1}{Tr_c} \quad (27)$$

$$T[i][j] = (1 - \rho) \times T[i][j] + \Sigma(\delta T[i][j]) \quad (28)$$

$$T[i][j] = (1 - \rho) \times T[i][j] + \Sigma \frac{1}{Tr_c} \quad (29)$$

where T is the pheromone matrix and Tr_c is the cost of the tour.

4.2.3 | Pareto optimal solution theory

In multi-objective optimization problems (MOPs), the various objectives slated for optimization frequently exhibit conflicting tendencies. As a result, these problems lack a singular, definitive optimal solution and cannot be resolved directly. Only by striving to optimize as many objectives as feasible can we arrive at a compromise solution set, often referred to as the Pareto optimal solution set. Subsequently, individuals can select their preferred solution from this set based on distinct requirements. In a broader context, no superior solution exists beyond the Pareto solution set; thus, it stands as the ultimate solution set. MOPs are typically representable using the following equation (30)³⁶:

$$F(x) = [f_1(x), f_2(x), \dots, f_m(x)] \quad (30)$$

$$x = (x_1, x_2, \dots, x_n) \in D$$

where x is a vector of n decision variables and $D \subset \mathbb{R}^n$ is the decision space. m is the number of objectives and $F(x) \in \Lambda \subset \mathbb{R}^m$ is the objective vector with m objectives. Λ is the objective space. The decision variables, denoted as x , play a pivotal role in this context. By selecting the correct value for x , we can attain the ideal solution to the optimization problem. It's essential to note that solutions within the Pareto solution set may not lead to the optimization of all objectives simultaneously. However, once one objective is fixed, the Pareto solution may enable the other objectives to reach their optimal values. Consequently, when seeking the optimal solution, our focus should be solely on the Pareto solution set. Pareto solutions exhibit a trait of compromising other objective functions while enhancing one, a characteristic that perfectly aligns with the attributes of MOPs. These Pareto solutions are solutions that do not dominate or outperform other alternatives, earning them the designation of non-dominated solutions. In the absence of any artificial criteria, they hold an

equivalent status within MOPs, often considered collectively as the Pareto frontier. If a solution satisfies Equations (31) and (32), and the goal function G applied to vector x' is less than or equal to the value of the goal function G applied to vector x , it qualifies as a pareto solution.³⁶

$$G(x') \leq G(x) \quad (31)$$

$$\text{Other Goals : } G(x') \geq G(x) \quad (32)$$

The Pareto dominance relation is established for the minimized multi-objective vector a comprising m objective components, applicable to any pair of decision variables $F(x) = [f_1(x), \dots, f_m(x)]$ and $x_1, x_2 \in X$, if and only if $i \in \{1, 2, \dots, m\}$, there is $f_i(x_1) < f_i(x_2)$, then x_1 dominates x_2 , which is recorded as $x_1 > x_2$. In the whole solution space S , the pareto solution set is a set composed of all the non-dominated solutions.²³ The pareto frontier is defined as follows: a series of non-dominated solutions form the Pareto solution set PS ,³⁷ and the surface composed of their corresponding vectors is pareto frontier (PF).³⁸

All constraints are transformed into convex, and the optimization problem can be reformulated as

$$\max S, C, f \sum_{i=1}^U -\alpha_i \times f_i \quad (33)$$

$$y_i \leq -\frac{1}{2}x_i^2, i = 1, 2, \dots, U \quad (34)$$

$$m_i \geq 2^{q_i} - 1 \quad (35)$$

$$q_i \geq \frac{d_o(1 - \sigma)}{B_i t_i} \quad (36)$$

Optimal results can be obtained by setting the initial value of y_i , $B_i t_i$, m_i , x_i updating variables, and performing iterations until the problem converges.

4.3 | PGA algorithm

Artificial intelligence (AI), machine learning (ML) Deep reinforcement learning (DRL) has recently become known as an essential technique for UAV trajectory planning in data gathering circumstances. However, considering the characteristics of the problem, using the PGA algorithm appears appropriate. The PGA algorithm emphasizes minimizing UAV travel distance, optimizing energy consumption, and achieving real-time data acquisition for enhanced situational awareness. The throughput capacity improvement and the combination of multi-UAV path planning, collision avoidance, and reliable communication with ground based IoT sensors are highlighted.

The PGA algorithm serves as the backbone of our multi-UAV path planning system, revolutionizing the efficiency and effectiveness of data acquisition and communication between ground based IoT sensors and UAVs. The algorithm is meticulously designed to seamlessly connect the PGA optimization process with the ISAC framework, ensuring a harmonious interplay between data collection, transmission, and real-time communication. It has been specially designed to maximize the collection of sensor data from ground based IoT sensors.

The proposed PGA algorithm for ISAC stands as a transformative solution that harmonizes data collection and communication in multi-UAV systems. By leveraging the power of PGA's optimization capabilities, the algorithm ensures that UAVs efficiently gather data from ground based IoT sensors while maintaining reliable communication links. This seamless integration of sensing and communication components empowers our system to achieve real-time data acquisition, enhanced situational awareness, and efficient monitoring of challenging terrains, underscoring its potential for transformative applications in diverse domains.

The process begins by setting the parameters for the PGA algorithm by initializing the population size, the number of generations, and determining crossover probabilities (p_{cross}) using Equation (24) and mutation probabilities (p_{mut}) using Equation (25) that influence the optimization process. The genetic algorithm component of the PGA is first utilized to determine the best ordering of the sensors to be visited. By evaluating different permutations of the sensors, the genetic algorithm identifies the best sequence to travel between the sensors. This process aims to minimize the overall distance traveled by the UAVs. The ant colony optimization mechanism is employed to refine the obtained sensor sequence and

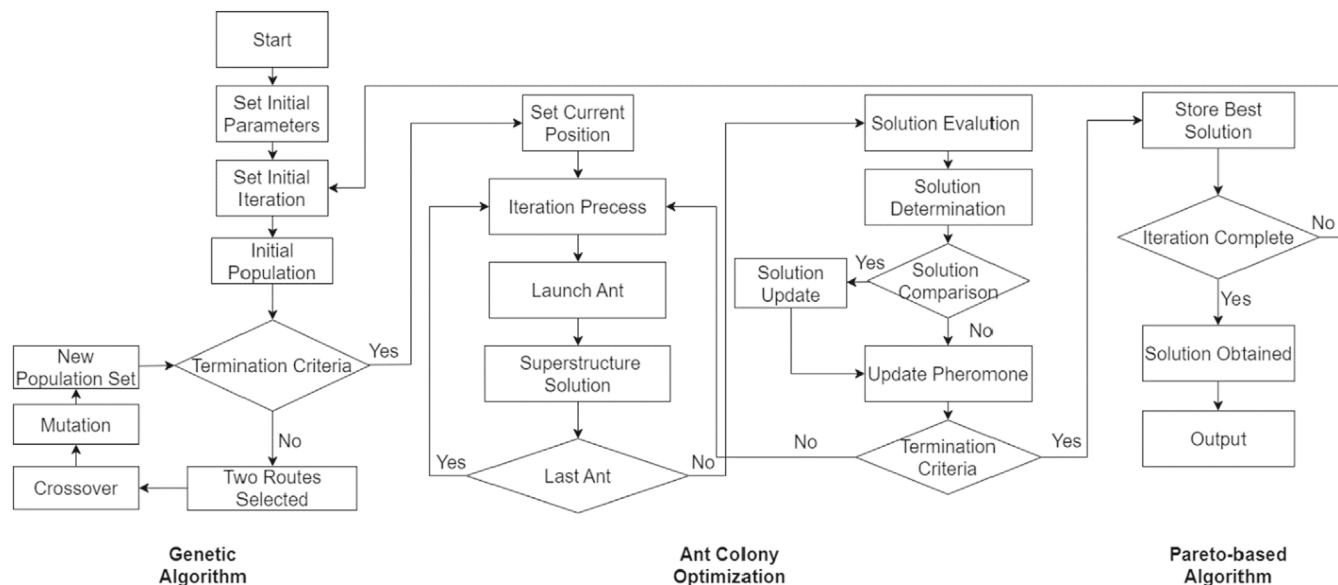


FIGURE 3 Multi UAV path planning using PGA Algorithm.

determine the optimal path between each pair of sensors and calculates the best trajectory between two sensors by computing the pheromone matrix with Equation (27), extending this calculation to all waypoints between sensor points. Multiple iterations are executed, generating various path copies, and computing multiple fitness functions, including the traveling fitness function ($FT_{fitness}$) using equations (2). Finally, the algorithm identifies the best $F(x)$ value from equation (30) by comparing it as described in equations (31) and (32). The ultimate selection of the best path is achieved through a combination of genetic algorithms (GA), ant colony optimization (ACO), multiple fitness functions, and Pareto solutions.

The integrated PGA algorithm discovers optimal or near-optimal paths for the UAVs. These paths aim to minimize the distance traveled and efficiently cover all the sensors in the sensing zones. Through the integration of the PGA algorithm, the multi-objective optimization process considers objectives such as distance traveled, sensor coverage, and energy consumption. The Pareto-based approach ensures that the resulting paths represent the trade-off between these objectives, providing a set of diverse and optimal solutions.³⁹ Multi UAV path planning using PGA algorithm is illustrated in Figure 3, which can maximize the collection of sensor data. The process begins with the UAV embarking on its journey, guided by an optimal path generated by the PGA algorithm. As the UAV approaches a sensor, it initiates a wake-up signal broadcast. Upon receiving this signal, the ground IoT sensor awakens and begins transmitting the previously collected environmental data to the UAV. The UAV acts as an intermediary, relaying this data to a designated base station before proceeding to the next sensor location. Notably, the UAV intelligently allocates bandwidth, initially splitting it evenly between sensing and communication tasks at 50%. However, it possesses the capability to dynamically adjust this allocation based on the real-time requirements, ensuring efficient and adaptable data collection and communication processes throughout the mission. This integrated approach enhances the overall effectiveness of data collection and transmission via UAVs in IoT sensor networks. Pseudocode for the PGA algorithm is provided in Algorithm 1.

Algorithm 1. Multi-UAV path planning using PGA algorithm

1: Input

Num of iterations: Number of iterations for the algorithm.

2: Num of ants: Number of ants to explore the search space

3: α : Weight parameter for pheromone levels

4: β : Weight parameter for heuristic information

5: ρ : Pheromone evaporation rate

6: Initial PP sequences: Initial paths for each UAV

7: S_{pb} : Sum of probabilities for pheromone updating

8: T : Pheromone matrix

- 9: H : Heuristic matrix
- 10: Output:
- 11: Initialize Initial PP sequences randomly;
- 12: Repeat for $t = 1$ to num of iterations:
- 13: Create m ants to explore the search space:
- 14: Current sensing location to each unvisited sensing area i :

$$\Sigma((T[C_i][i])^\alpha) \times (H[C_i][i]^\beta)$$

- 15: For each unvisited sensing area i :
- 16: $pb[i] = \frac{((T[C_i][i])^\alpha) \times (H[C_i][i]^\beta)}{S_{pb}}$
- 17: Update Tr_c
- 18: $\delta T[i][j] = \frac{1}{Tr_c}$
- 19: $T[i][j] = (1 - \rho) \times T[i][j] + \Sigma(\delta T[i][j])$ for all ants
- 20: Apply Pheromone Evaporation:
- 21: Update $T[i][j]$ as follows: $T[i][j] = (1 - \rho) \times T[i][j]$
- 22: Return Tr_c
- 23: $pb[i] = \frac{T[S_s][i]^\alpha \times H[S_s][i]^\beta}{\Sigma(T[S_s][i]^\alpha \times H[S_s][i]^\beta)}$ for all unvisited sensors j
- 24: Update Pareto Frontier
- Update T based on Pareto Frontier
- 25: $T[i][j] = (1 - \rho) \times T[i][j] + \Sigma \frac{1}{Tr_c}$
- $FT_{fitness} = FT_{obj} + FT_{const}$ for all tours in Pareto Frontier.
- 26: Perform update on T : $T[i][j] = (1 - \rho) \times T[i][j]$
- 27: Return Pareto Solution
- 28: End

The optimized paths obtained through the PGA algorithm ensure that the UAVs traverse the sensors in the most effective manner, reducing redundant movements and enhancing overall system performance. The pseudocode of PGA algorithm is given in Algorithm 1, this can maximize sensor data acquisition.

Farther more for wake-up radio signal (WUR) we need to consider low power consumption, high sensitivity, resistance to interference and fast wake-up.⁴⁰ To gain insight into the minimum necessary sensitivity, the Friis transmission equation comes into play. The Friis transmission equation establishes an equality between the power at the terminals of a receiving antenna and the product of the power density of the incoming wave and the effective aperture of the receiving antenna, all within idealized conditions. This equation is applicable when considering another antenna transmitting a known amount of power at some distance away. The received power (P_r) is expressed as follows:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^n} \quad (37)$$

in this equation, P_t represents the transmitted power, with G_t and G_r denoting the antenna gain on the transmitter and receiver sides, respectively. The variable d signifies the transmission distance, λ corresponds to the wavelength of the utilized frequency, and n represents the path loss exponent.

5 | RESOURCE ALLOCATION FOR ISAC

In integrated sensing and communication systems, resource allocation is a critical issue that determines the performance and efficiency of both the sensing and communication tasks.⁴¹ These systems aim to share the same hardware and spectral resources to carry out both sensing and communication activities.³² Advanced algorithms consider both tasks simultaneously, allocating bandwidth in a way that maximizes the overall system performance. Typically, the goal is to maximize some measure of system performance. This could be spectral efficiency, signal-to-noise ratio

(SNR), or a composite metric that considers both sensing and communication tasks. Mathematically, this could be expressed as:

$$\text{maximize } f(x) \quad (38)$$

where $f(x)$ is the objective function dependent on the variable x representing the allocation vector. A simple objective function $f(x_c, x_s)$ could aim to maximize the sum of some utility functions $U_c(x_c)$ and $U_s(x_s)$ that represent the quality or effectiveness of the communication and sensing tasks, respectively.

$$\text{maximize } f(x_c, x_s) = U_c(x_c) + U_s(x_s) \quad (39)$$

The utility functions $U_c(x_c)$ and $U_s(x_s)$ might depend on various factors such as data rate, signal-to-noise ratio, or any other performance metric. Which can be calculate as:

$$U_c(x_c) = \log(1 + \text{SNR}_c \times x_c) \quad (40)$$

$$U_s(x_s) = \log(1 + \text{SNR}_s \times x_s) \quad (41)$$

Let us consider to be the bandwidth allocated to the sensing task T as the time allocated for sensing, as the sampling rate, and N as the total number of samples collected. Number of samples collected the total number of samples N can be given as:

$$N = T \times R_s \quad (42)$$

The data collection rate D_s for the sensing task, often measured in bits per second (bps) or similar units, might be given by:

$$D_s = x_s \times \log_2(1 + \text{SINR}_s) \quad (43)$$

Here, SINR_s is the signal-to-noise ratio for the sensing task. The total data D_t collected over the time T could be expressed as:

$$D_t = D_s \times T \quad (44)$$

$$D_t = x_s \times \log_2(1 + \text{SINR}_s) \times T \quad (45)$$

where we want to maximize the D_t and $X_c + X_s \leq B$. Convex optimization techniques can be highly effective for solving resource allocation problems, including the bandwidth allocation in integrated sensing and communication systems. Such techniques aim to find the optimal solution within a convex set, often using algorithms that guarantee convergence to a global minimum. In this scenario, the objective is to allocate bandwidth in a manner that maximizes the overall system performance subject to various constraints like Quality of Service (QoS), latency, and spectral efficiency.

ISAC resource allocation using convex optimization resource allocation in Algorithm 2 begins by initializing sensing (S) and communication (C).⁴² The process involves iterations, aiming for optimal resource allocation S and C. The objective is to maximize an f function by summing user-specific weights α_i . The algorithm repeats until the objective converges, using a mathematical expression. The resource allocation concludes when the process ends.

Algorithm 2. ISAC resource allocation using convex optimization resource allocation algorithm

- 1: Input
- U : Number of users.
- 2: α_i : Weight parameter for user i
- 3: f_i : Utility function for user i

- 4: S : Sensing matrix for resource allocation
- 5: C : Communication matrix for resource allocation
- 6: Output:
 - Optimal Resource Allocation (S, C).
- 7: Initialize sensing and communication for resource allocation S and C .
- 8: Repeat for optimal resource allocation S and C

$$\max S, C, f \sum_{i=1}^U -\alpha_i \times f_i$$

- 9: Until the objective value converges

$$\max S, C \sum_{i=1}^U e^{-\frac{1}{2}m \left[\frac{N_0 B_l \left[2 \left[\frac{d_0(1-\alpha_i)}{B_l I_0} \right] - 1 \right]}{\delta P_l} \right]^2} \times \eta(o_i)$$

- 10: End

6 | SIMULATION RESULTS AND DISCUSSION

Within this paper, we have developed an operational environment using MATLAB version R2023a. This environment serves as the platform for assessing the operational performance of the PGA algorithm, specifically tailored for multi-UAV path planning within ground based IoT sensor networks associated with ISAC. The simulation encompasses various critical parameters that collectively shape the behavior and performance of the wireless sensor network and UAVs. The simulation takes place within a cell area of 30×30 . In this context, IoT sensors, numbering 30, are integral to the network's data collection mechanism. The sensing range of UAV is up to 30 m. The simulation involves three UAVs that have the flexibility to fly at altitudes ranging from 100 to 300 meters, lending versatility to their operational profiles. The simulation entails 250 iterations for each UAV, facilitating in-depth analysis of network dynamics. The optimization strategies are underpinned by the GA and ACO approach. Within the GA framework, parameters such as selection probability (ranging from 0.02 to 0.1), mutation probability (fixed at 0.08), and population size (varying from 10 to 30) influence the optimization process. The ACO methodology incorporates a heuristic factor of 1 and deploys a variable number of ants (10 to 30) to enhance network efficiency. Wireless communication parameters include a transmit power of 23 dBm, a carrier frequency of 3.75 kHz, and an available bandwidth of 180 kHz. Sensors are positioned at a height of 1.5 meters, optimizing their line-of-sight and data collection capabilities. A shadow fading standard deviation of 4 dB accounts for signal variation due to environmental conditions. Notably, transmitted data size varies between 1 kilobyte and 2 megabytes, accommodating diverse data types and sizes in the simulation. The main simulation parameters are listed in Table 1.

Figure 4A, we observe the operational environment with randomly distributed ground based IoT sensors. This setup represents the real-world scenario in which the sensors collect valuable data across the sensing zones. However, the challenge lies in efficiently collecting and transmitting this data while optimizing resource allocation.

Figure 4B,C present the optimized paths for three UAVs, namely UAV1 marked as red, UAV2 marked as green, and UAV3 marked as blue, which is generated by the PGA algorithm. These paths vividly demonstrate the prowess of the PGA Algorithm in efficiently allocating resources for data collection and communication. Figure 4B,C showcase the 3D view of the planned paths generated by the PGA Algorithm. This path optimization process incorporates multiple objectives such as minimizing travel distance, maximizing sensor coverage, optimizing energy consumption, and accounting for communication constraints. The algorithm's blend of GA and ACO ensures that the UAVs traverse the most efficient paths, minimizing energy consumption and optimizing data collection while adhering to communication range limitations.

In Figure 5A of our study, we present a comprehensive analysis of Unmanned Aerial Vehicle (UAV) energy consumption in relation to the number of cells covered. The x-axis denotes the varying quantity of cells, ranging from an initial coverage of approximately 45 cells. Concurrently, the y-axis illustrates the corresponding energy consumption, spanning from 0 to approximately 20 units. This depiction reveals a discernible trend showcasing the intricate interplay between UAV energy utilization and the extent of cellular coverage.

TABLE 1 System-level simulation parameters.

Parameters	Value
Cell area sensor network	30×30
Number of UAVs	3
Number of IoT sensor	30
Sensor type	5 ~ 8
Selection probability (GA)	0.02 ~ 0.1
Mutation probability (GA)	0.08
Population size (GA)	10 ~ 30
Heuristic factor (ACO)	1
Number of ants (ACO)	10 ~ 30
Sensor type	CAT-M1 IOT
Number of sensors (For each UAV)	10
Sensing range	30 m
UAV altitude	100 ~ 300 m
Iterations per UAV	250
Transmit power	23 dBm
Bandwidth	1.4 MHz
Height of sensors	1.5 m
Shadow fading standard deviation	4 dB
Transmitted data size	1 KB–2 MB

Figure 5B extends this investigation by examining energy consumption concerning the number of sensors employed. The x-axis delineates the quantity of sensors, consistently maintained at 10 throughout the experimental setup. The y-axis articulates the corresponding energy consumption, varying from 0 to around 20 units. Notably, each incremental addition of a sensor results in a gradual augmentation of energy expenditure. Strikingly, the parallels between Figure 5A,B underscore a noteworthy consistency in the observed energy consumption patterns.

In Figure 6, we conducted a comprehensive comparison against the conventional path planning architecture. We evaluated the utility values of the optimal path using three approaches: the A* algorithm,⁴³ the A* + GA algorithm, and ACO + GA. These methods were all equipped with the proposed utility function. In the case of the A* + GA algorithm, we integrated the suggested crossover and mutation techniques. Additionally, we incorporated the cell selection probability conditions from the ACO + GA algorithm.⁴⁴ Notably, across a range of cell selection probability conditions, the PGA method we propose consistently outperforms the others, demonstrating the highest and most favorable utility values.

Figure 7A focuses on the concept of Signal-to-Interference-plus-Noise Ratio (SINR) intersection points, which hold valuable insights into communication performance. These intersection points mark the locations where SINR values overlap. They represent areas where communication quality achieves a balance, signaling optimal resource allocation and minimized interference. In Figure 7B, we focus on the SINR to investigate the communication dynamics between UAVs and ground based IoT sensors. The figure showcases a 3D representation where each point corresponds to a specific location within the operational area. The SINR value at each point is color-coded to provide a visual representation of communication quality. In this visualization, the color spectrum indicates the range of SINR values. Dark blue hues correspond to SINR values lower than -6 dB, indicating areas with relatively weaker communication quality due to higher interference or noise levels. On the other hand, brighter colors such as yellow indicate SINR values greater than 8 dB, representing regions with robust communication quality.

In Figure 8A The sensor bars that are closer to the maximum throughput value ($6e7$) are those that have been allocated optimal resources using the convex optimization resource allocation technique with UAVs path planning PGA algorithm. These sensors can transmit data at the highest possible rate. Sensors with throughput values that are closer to the minimum throughput value ($4.5e7$) might not have received as favorable resource allocation. Consequently, their

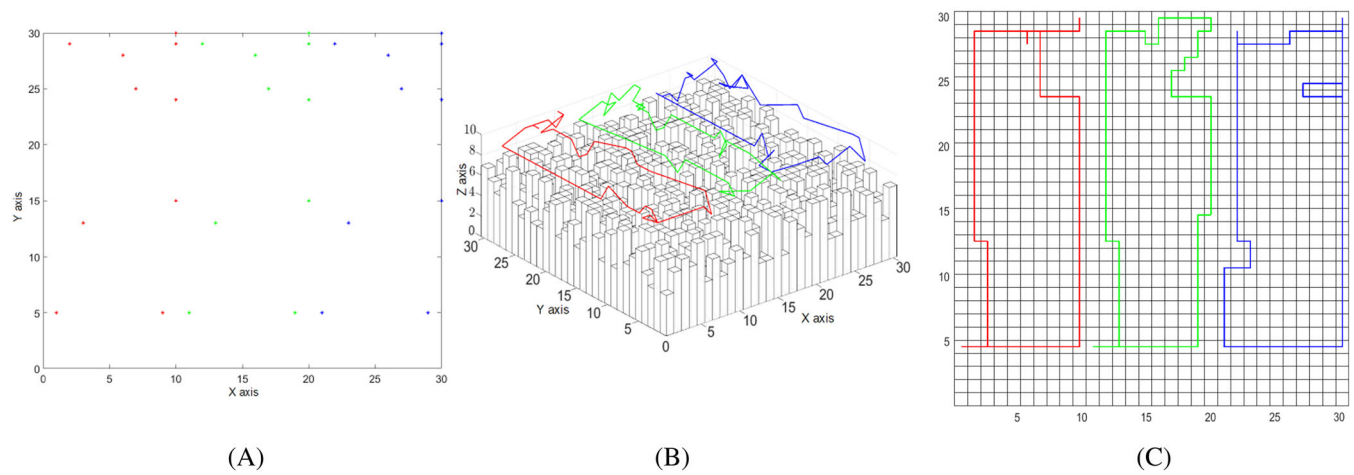


FIGURE 4 Sensing area with generated path planning using PGA algorithm: (A) Randomly distributed sensors; (B) Generated path 3D view; (C) Generated path top view.

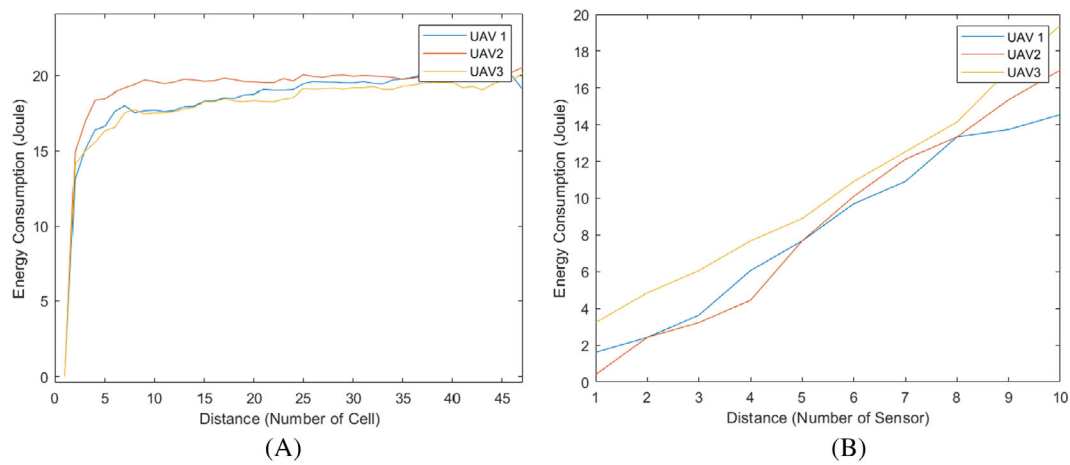


FIGURE 5 Multi UAV energy consumption (A) according to the covered cell distance (B) according to the number of sensors.

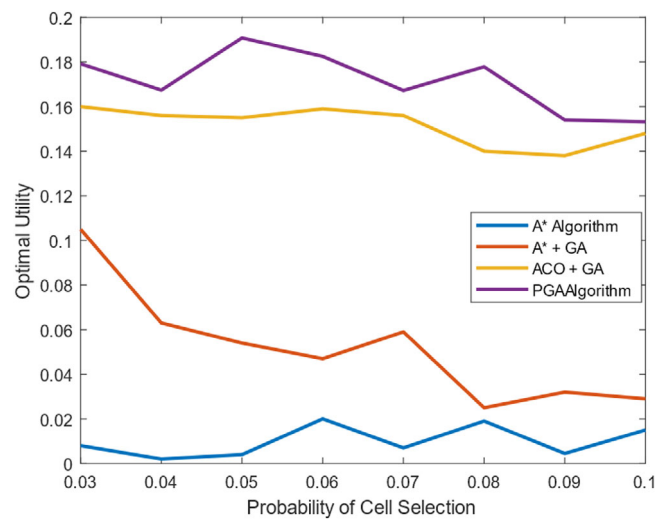


FIGURE 6 Comparative analysis of the proposed method against conventional algorithms.

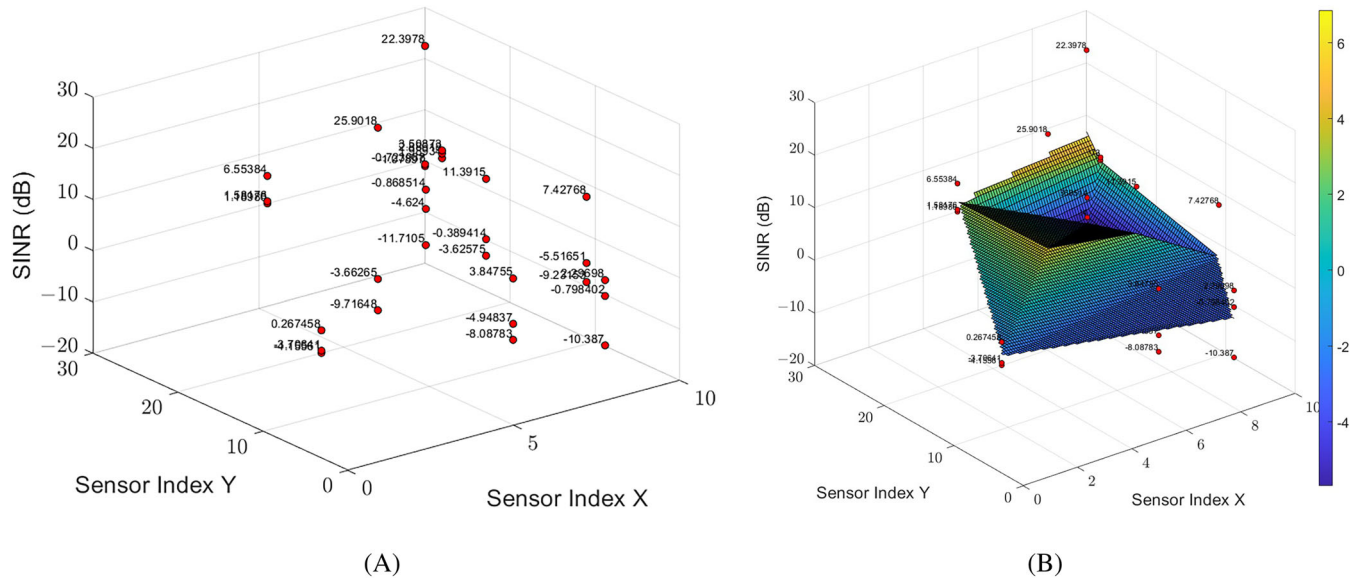


FIGURE 7 IoT-based sensor data acquisition for UAV communication (A) Intersection points of sensor nodes and (B) 3D representation of SINR (dB) by sensor node.

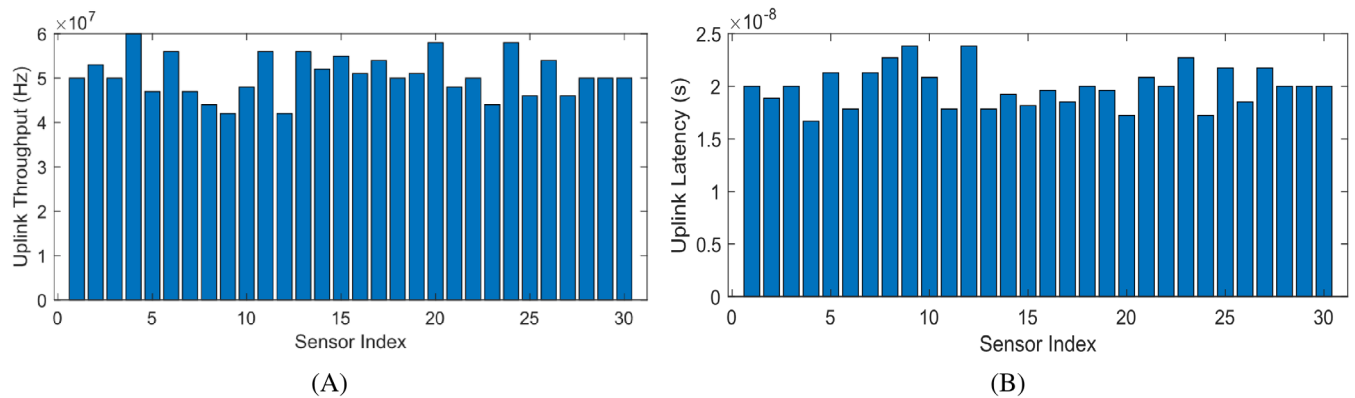


FIGURE 8 Performance of the ISAC system in IoT-based multi-UAV uplink communication (A) Throughput and (B) Latency.

data transmission rate is slightly lower. Sensors with throughput values falling in between the maximum and minimum values represent a range of intermediate resource allocations. These sensors achieve throughput values that correspond to their specific resource allocations. Moving to Figure 8B examines the effect of sensor count on latency in the ISAC system. Notably, the maximum latency remains 240 microseconds, while the minimum latency impressively stayed around 150 microseconds. We found the average latency is 195 microseconds.

Figure 9 effectively visualizes the execution time dynamics for individual sensors, highlighting the efficiency of the combined sensing and communication processes of ISAC. The unveils outcomes from our investigation into resource allocation's impact on uplink throughput in UAV-to-ground sensor communication. Figure 9A illustrates uplink throughput comparisons with and without resource allocation. The with resource allocation curve in Figure 9A showcases uplink throughput under effective resource allocation strategies. It spans from $6.5e7$ at its peak to $4.2e7$ at its lowest, revealing significant enhancements in performance. In contrast, the without resource allocation curve presents considerably lower uplink throughput values reaching a maximum of $.45e7$ and a minimum of $.4e7$. These findings underscore resource allocation's pivotal role in elevating communication efficiency between UAVs and ground based IoT sensors.

Figure 9B illustrates a comparative performance of uplink latency in IoT and UAV communication and sensing performance of ISAC, considering scenarios with and without resource allocation. The graph depicts uplink latency curves, reflecting the time delay for data transmission. When resource allocation is implemented, uplink latency ranges from

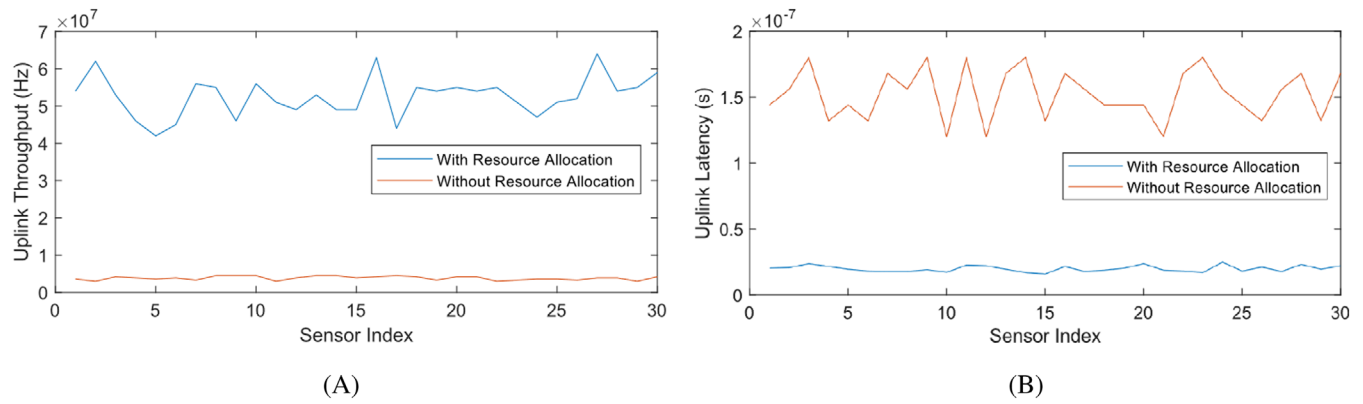


FIGURE 9 ISAC resource allocation performance under IoT-based multi-UAV uplink communication (A) Throughput (B) Latency.

$1.75\text{e-}7$ s (maximum) to $1.2\text{e-}7$ s (minimum), indicating the benefits of effective resource distribution. Conversely, in the absence of resource allocation, uplink latency exhibits higher values, varying from $.22\text{e-}7$ s (maximum) to $.15\text{e-}7$ s (minimum). Notably, the graph includes 30 sensor indexes, each representing individual sensors. These indexes provide insight into uplink latency for UAV-to-ground sensor communication, offering a nuanced understanding of how resource allocation impacts communication and sensing performance. This analysis underscores the importance of resource allocation in reducing uplink latency and enhancing overall system efficiency in IoT-enabled UAV deployments.

7 | CONCLUSIONS

This research describes multi-UAV path planning for ISAC in ground-based IoT sensor networks, in which we built connection between UAV and ground-based IoT sensor. The PGA algorithm's pareto-based optimization and genetic ACO combination effectively optimized UAV trajectories in complicated situations, ensuring efficient data collecting, collision avoidance, and reliable communication. We demonstrated multi-UAV path planning through simulations, as well as integrated sensing and communication between UAVs and ground based IoT devices. The PGA algorithm provides an ideal path, which UAV uses to perceive and communicate with ground based IOT sensors. The data is collected by the UAV from the ground based IOT sensor and transmitted to the base station through 5G mobile connection. Our system has a maximum throughput of roughly 95% and a maximum latency of 17.50 microseconds. In conjunction with resource allocation, our suggested UAV path planning PGA algorithm is intended to maximize the efficiency of ground-based sensor data collecting. Examining the PGA Algorithm's scalability and adaptation to broader and dynamically changing situations increases ISAC's potential. Improvements to communication protocols and procedures have the potential to improve data transmission efficiency, particularly in congested or interference-prone areas, hence enhancing ISAC capabilities.

The incorporation of machine learning and artificial intelligence techniques has the potential to upgrade the ISAC concept, enabling more intelligent and autonomous UAV path planning. Collaborative multi-UAV missions directed by swarm intelligence principles offer intriguing promises for increasing ISAC's efficiency and efficacy. Future initiatives should involve thorough field experiments and real-world deployments under varied settings to validate and develop the ISAC concept. This practical validation will pave the path for the broad use of integrated multi-UAV sensing and communication technology in a variety of industries and applications. We foresee ongoing breakthroughs by actively researching these future avenues with ISAC at its center, sparking the realization of efficient, adaptive, and socially beneficial multi-UAV systems.

AUTHOR CONTRIBUTIONS

Supervision and investigation: KyungHi Chang; Rahman Md Azadur and Chaitali J. Pawase are contributed equally in this paper for methodology and writing original draft preparation; Writing-review and editing: KyungHi Chang, Rahman Md Azadur and Chaitali J. Pawase. All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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